

# Worksheet 3

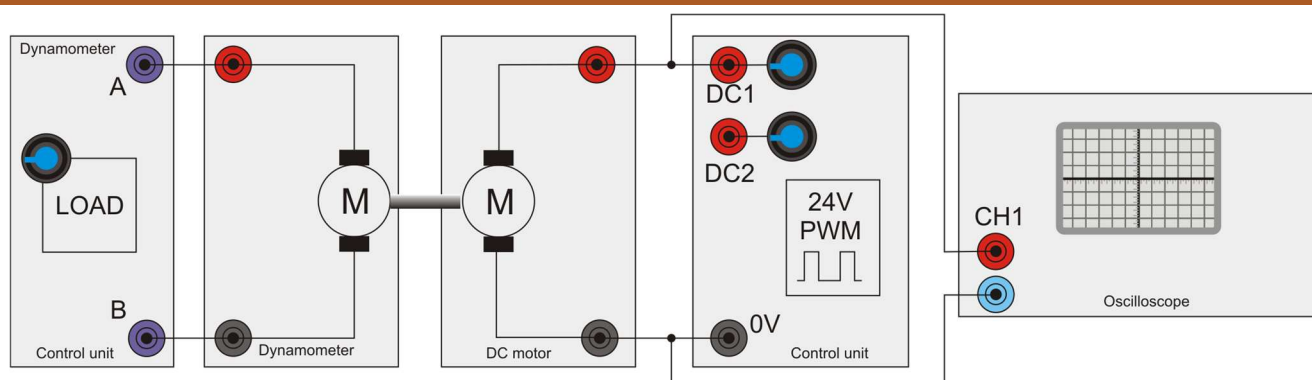
## Understanding the DC power supply

## Modern electrical machines



The permanent magnets occupy a small space in a DC motor compared to the space needed for the windings in motors where the magnetic field is supplied by current-carrying coils. As a result, DC motors can be physically quite small for their power rating.

The food blender in the photograph uses a DC motor.



A motor can be controlled by varying the DC voltage, and hence the power, supplied to it.

### Over to you

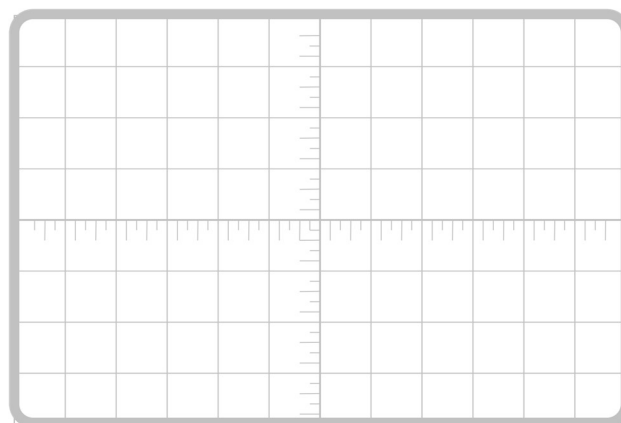
- 1) Adjust the DC1 potentiometer so that it is in about a quarter way of the way round.
- 2) Use an external oscilloscope, not a PC-based one, to look at the signal from DC1.
- 3) Use the upper graph opposite to sketch the waveform of this signal.
- 4) Turn the DC1 potentiometer about three-quarters of the way round and draw the resulting waveform on the lower graph.

### So what?

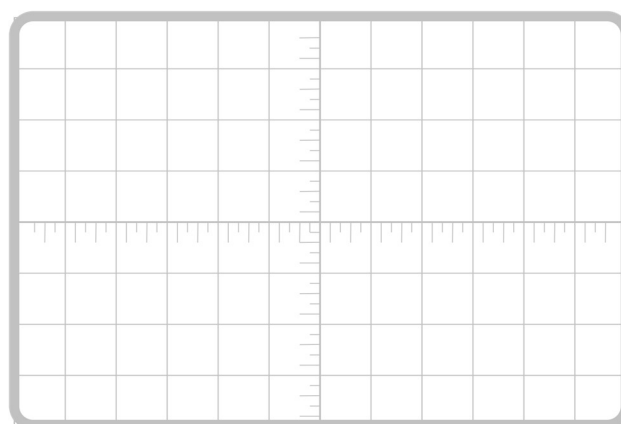
Modern DC motor power supplies do not vary the output in an analogue fashion, but use pulse-width modulation or PWM. In this, the mark : space ratio of the output is varied.

In the control unit, a bank of FET's deliver a PWM signal generated by a computer. This can be used with DC, with single-phase and with three-phase motors.

**Do not use a PC-based oscilloscope.**



DC1 - one-quarter max.



DC1 - three-quarters max.